ROOT CAUSE ANALYSIS

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Management Plane Loss After MM Host Rebuild

Root cause analysis of vSAN object inaccessibility cascade triggered by Ensure Accessibility selection during ESXi host reimage, resulting in vCenter, SDDC Manager, and Lifecycle Manager loss.

VSAN | MAINTENANCE MODE | HOST REBUILD | OBJECT INACCESSIBILITY | RCA

VCF 9.0 VMware Cloud Foundation

VCF 9.0 Management Plane Loss — Root Cause Analysis

Prepared by: Virtual Control LLC **Date:** May 2, 2026 **Document Version:** 4.1 **Classification:** Internal — Lab Environment **Incident Date:** 2026-05-01 / 2026-05-02 **Severity:** P1 — Total management plane loss; full redeploy required

> **PowerShell users:** Every bash block in this document that targets a workstation-runnable command (local Windows tooling, REST API client) has a paste-ready PowerShell sibling block underneath it. For the full substitution table — `curl` → `Invoke-RestMethod`, base64 auth headers, here-docs → here-strings, `jq` / `python -m json.tool` → `ConvertFrom-Json` / `ConvertTo-Json -Depth N`, `plink` → `Invoke-SSHCommand` / `ssh.exe`, etc. — see POWERSHELL-TRANSLATION-REFERENCE.md. Blocks targeting the ESXi shell only (`esxcli`, `vsish`, `objtool`, `/etc/init.d/*`) or the VCSA appliance shell only (`vecs-cli`, `service-control`, `psql`, `systemctl`, `journalctl`) are flagged "Run on the ESXi host only" / "Run on the appliance only" with no PS sibling — they cannot run from a Windows workstation.

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Related RCAs in this library (`06-Troubleshooting/`): - `NSX-Manager-Cold-Start-Troubleshooting-Report` — NSX Manager appliance recovery (4 addenda: Corfu DB corruption, btrfs read-error after power-off, restore-from-backup path, post-restore stuck-state → fresh redeploy). Addendum D's L-32 cites this doc as the source of the FDM-only rule. - `Fleet-Operations-Integration-Recovery-RCA` — VCF Operations ↔ Fleet Management half-broken integration registration; KB-aligned repair via Disconnect/Reconnect with DB-level cleanup.

1. Executive Summary

On 2026-05-01 a planned ESXi host reimage of nested host **esxi02** in a 4-host VCF 9.0 lab cluster resulted in total loss of the management plane: **vCenter Server, SDDC Manager, and Lifecycle Manager** all became unrecoverable. Nine vSAN objects entered the *inaccessible — APD (All Paths Down)* state and could not be repaired.

The proximate cause was the selection of **Ensure Accessibility** when entering vSphere Maintenance Mode prior to the host reimage, instead of the correct option for a destructive rebuild — **Full Data Migration**. EA does not evacuate vSAN components from the host's local disks; when the disks were subsequently wiped, every object that retained a stripe of single-component placement on esxi02 lost its only copy of those bytes. With no surviving replica for those stripes, FTT=1 RAID-1 was insufficient to maintain availability.

The cascade then accelerated: the simultaneous compliance-rebuild and rebalance load on the cluster's three remaining HDD-backed nested hosts produced sufficient I/O latency to abort the EXT4 journal on the vCenter VM's database disk, taking vCenter offline mid-event. Without vCenter, no `Repair Cluster Objects` workflow was available; no file-based backup of vCenter existed.

A non-destructive recovery attempt (DOM ownership abdication per Broadcom KB 405527) successfully transferred ownership but did not restore object accessibility, confirming the missing components were not recoverable from the cluster's surviving members. The 9 dead objects were cleaned with `objtool delete -f` per KB 428195, and a fresh management plane is being redeployed.

The single most important corrective control going forward: **Maintenance Mode for any destructive host rebuild MUST be entered with Full Data Migration**. Ensure Accessibility is for transient outages with rapid host return, not for reimage events.

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2. Environment Reference

2.1 Cluster Topology

Host	FQDN	IP	Role	State at incident
esxi01	esxi01.lab.local	192.168.1.74	Compute / vSAN	Healthy throughout
esxi02	esxi02.lab.local	192.168.1.75	Compute / vSAN	Reimaged — vSAN data destroyed
esxi03	esxi03.lab.local	192.168.1.76	Compute / vSAN	Healthy; became master post-event
esxi04	esxi04.lab.local	192.168.1.82	Compute / vSAN	Healthy throughout
vCenter	vcenter.lab.local	192.168.1.69	Management	EXT4 journal abort → unrecoverable

2.2 Storage Architecture

Item	Value
vSAN architecture	OSA (Original Storage Architecture)
vSAN sub-cluster UUID	`52b68472-c149-d706-5c7e-ef39e7136ffa`
Storage policy default	FTT=1 RAID-1
Backing media	Nested ESXi VMDKs on host-local HDDs (HGST SATA + NVMe in migration)
Cluster member count, healthy baseline	4
Cluster member count, post-incident	3 (esxi02 absent from vSAN sub-cluster)

2.3 Management VMs Affected

VM	Inaccessible vSAN Objects	Outcome
vCenter Server (vcenter.lab.local)	Namespace + snapshot delta vmdk	EXT4 journal abort during cascade; unrecoverable, redeploy required
SDDC Manager	VMDK 2 (17 GB) + VMDK 3 (133 GB)	Unrecoverable, redeploy required
Lifecycle Manager (LCM)	VMDK 1 (12 GB)	Unrecoverable, redeploy required
`.vsan.trace` system object	namespace	System object — non-critical, regenerated
2 unidentified objects (Type N/A)	metadata corrupt	Cleaned with objtool
1 small orphan vdisk (0.07 GB)	unknown	Cleaned with objtool

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3. Problem Statement

3.1 Symptom Description

After completing a fresh ESXi reimage of esxi02 and rejoining it to the cluster, the following symptoms manifested in sequence:

1. vSphere Skyline Health flagged 9 objects in `inaccessible` state with `Lost data availability.(APD)`.
2. SDDC Manager and LCM VMs displayed as **Orphaned** in vCenter inventory.
3. vCenter VM became increasingly unresponsive; the VM console showed cascading EXT4 errors on `/dev/dm-14: EXT4-fs error (device dm-14) ext4_journal_check_start:83: Detected aborted journal EXT4-fs error (device dm-14): failed to convert unwritten extents to written extents -- potential data loss! (inode 674342, error -30)`
4. vCenter UI ceased responding; subsequent power-on attempts left the VM stuck in *Invalid* state in the host inventory.

3.2 Affected Components

Layer	Component	State
vSphere	vCenter Server 9.0.1	Powered off, EXT4 RO, services unstartable
VCF	SDDC Manager	Orphaned then objects inaccessible
VCF	Lifecycle Manager (LCM)	Orphaned then objects inaccessible
Storage	vSAN datastore	9 of 111 objects inaccessible (102 healthy)
Cluster	vSAN sub-cluster	Reduced to 3 members; esxi02 absent

3.3 Business Impact

- Total management plane outage on the affected workload domain.
- No automated failover possible; manual redeploy required.
- Estimated end-to-end recovery: 1 working day (VCSA redeploy + cluster re-onboard + SDDC Manager redeploy when scheduled).
- Customer impact in production-equivalent context would have been P1 with potential SLA breach.
- In this lab, the impact was contracted-work delay due to time spent on recovery.

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4. Root Cause Analysis

4.1 RCA Summary

Root cause: Selection of *Ensure Accessibility* maintenance-mode option prior to a destructive ESXi reimage of a vSAN cluster member, when *Full Data Migration* was the correct option for the operation type. EA preserves I/O availability by lazily rebuilding components only after a 60-minute repair-delay timer; it does not evacuate components before the host enters MM. When the host's disks were subsequently wiped, vSAN objects that depended on components on those disks (specifically RAID_0 stripes whose only copy resided on esxi02) lost data availability permanently.

4.2 The Maintenance Mode Decision

vSphere offers three modes for entering Maintenance Mode on a vSAN host:

Mode	Behavior	Correct Use Case
Full Data Migration	All vSAN components on this host are evacuated to other hosts before MM completes. Slow but state is fully steady when the host departs.	**Host reimage, hardware replacement, decommission**
Ensure Accessibility	Only components whose loss would render objects inaccessible are migrated. Components covered by FTT remain on the host.	Brief outages where the host returns within the 60-min repair delay timer
No Data Migration	No components migrated. Cluster relies on existing FTT alone for the duration of host absence.	Quick reboots only

Per the user's own *VCF 9.0 ESXi Host, vCenter & NSX Manager Rebuild Plan* (April 25, 2026, Section 4 Preconditions):

> *"Preserve vSAN disk partitions during ESXi reinstallation. Failure to preserve partitions forces a full disk-group rebuild and extended resync (KB 327032)."*

Reimaging esxi02 with a fresh ESXi install constituted disk-group destruction. The correct MM option was *Full Data Migration* — the host's components needed to be evacuated *before* the disks were destroyed, not deferred-rebuilt afterward.

4.3 Why Ensure Accessibility Caused the Cascade

When EA is selected and the host then has its disks wiped:

1. **Components on esxi02's disks become permanently absent.** The cluster's `Cluster-wide Object Repair Timer` (default 60 min, KB 327031) starts. After timeout, vSAN initiates compliance rebuilds for all affected objects.
2. **All affected objects rebuild simultaneously.** With ~110 objects in the cluster and many having components on the absent host, the cluster initiated a large parallel rebuild workload.
3. **Rebuild I/O exceeded HDD-backed nested vSAN throughput capacity.** Sustained ~115 MB/s reads were observed across the surviving members; this is the physical ceiling of the underlying media. With the rebuild workload competing with foreground guest I/O, latency spiked.
4. **Some objects had RAID_0 stripes with only single-component placement on esxi02.** With no surviving mirror copy of those specific stripes, the entire object became unreadable — quorum loss at the stripe level cannot be repaired by the cluster's redundancy scheme alone.

A representative case from forensic inspection (`.vsan.trace` object `b3a68669-ad9f-8094-f6d8-000c29649d0a`):

```
RAID_1
 Component: 3a5cee69-4364-...   ABSENT  (was on esxi02 disk)
 RAID_0
   RAID_1
      Component b58ef369-e39e (ACTIVE, esxi03)
      Component 7650f569-b77c (ABSENT)
   Component 8c16f169-49bf      ABSENT  ← single-component stripe, no mirror
   [...further stripes mostly ACTIVE on esxi04...]
 Component 7650f569-f38b        RECONFIGURING (stalled, 16 GB pending)
```

The single-component ABSENT stripe (`8c16f169-49bf`) makes the RAID_0 unreadable; the top-level RAID_1's only remaining leg is RECONFIGURING and therefore not yet serving reads. Object inaccessible.

4.4 The vCenter EXT4 Journal Abort

vCenter Server Appliance runs Photon OS with EXT4 filesystems. Its database (PostgreSQL) is on a dedicated vmdk mapped via LVM to `/dev/dm-14`. During the cluster-wide rebuild storm, sustained write latency exceeded EXT4's journal commit timeout. When the journal cannot be committed, EXT4 takes the protective action of aborting the journal and remounting the filesystem read-only:

```
EXT4-fs error (device dm-14): Detected aborted journal
EXT4-fs error (device dm-14): failed to convert unwritten extents to written
  extents -- potential data loss!
```

With the database disk RO, vCenter's PostgreSQL service began failing writes. vmware-vpostgres health degraded; dependent services (vpxd, sps, content library) followed. The user attempted a graceful shutdown which was blocked by the failed services; a hard power-off was eventually required.

Subsequent power-on did not recover the appliance — the EXT4 journal was sufficiently damaged that PostgreSQL could not initialize cleanly. With no file-based backup configured, this state is not recoverable in place.

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5. Contributing Factors

5.1 Incorrect AI Assistant Guidance

When the user asked an AI assistant (session ID `46f1b37d-9f39-41ea-b592-32cd05b616f8`, message timestamp **2026-05-02 00:13:13 UTC**) which of two MM options to choose between *Ensure Accessibility* and *No Data Migration*, the assistant recommended **Ensure Accessibility**. The assistant noted within the same response that *Full Data Migration* was the technically correct answer for a rebuild scenario, but did not refuse the false binary or insist on the third option — it recommended the lesser of two wrong choices.

This recommendation directly contradicted the user's own published *VCF 9.0 ESXi Host, vCenter & NSX Manager Rebuild Plan* (April 25, 2026), which mandates Full Data Migration with citations to Broadcom KB 391111 and KB 327032. The user followed the assistant's recommendation in the moment of the operation rather than re-checking the documented plan.

5.2 Bypass of SDDC Manager Workflow

The host reimage was performed by entering MM and reinstalling ESXi directly, rather than via SDDC Manager's documented *Decommission Host → Reinstall → Commission Host* workflow. The SDDC Manager flow includes a coordinated VCF-state evacuation that performs the equivalent of Full Data Migration before allowing the host to leave the workload domain. Bypassing this control removed a safety check that would have prevented the EA selection from being possible.

5.3 No File-Based Backup of vCenter

vCenter Server 9.0 supports native file-based backup to SFTP/FTPS/HTTP(S)/NFS/SMB/SCP targets, configurable via VAMI (port 5480). At incident time, vCenter had no backup target configured. When the journal abort rendered the appliance unrecoverable, the only remaining option was full redeploy, with full loss of:

- VC inventory database (alarms, permissions, custom attributes)
- VDS port allocation history
- License assignment records
- DRS/HA history and rules
- Tags and categories

Per Broadcom KB 428053, in VCF 9.0 license application requires VCF Operations to be reachable. Without vCenter, the path to license a freshly-deployed replacement also depended on whether VCF Operations was operational — adding a second-order dependency.

5.4 vCenter Running with Active Snapshot on Affected vSAN

The vCenter VM had an active snapshot at the time of the rebuild event. The snapshot delta vmdk (`vcenter-000001.vmdk`) was placed on the same vSAN datastore that became unstable during the cascade. When the namespace object went inaccessible, the running snapshot delta also became unreachable — compounding the recovery difficulty (even if the base disk had remained healthy, the snapshot chain could not be consolidated).

Best practice is to remove all active snapshots before any storage-impacting operation.

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6. Timeline of Events

All times in UTC unless otherwise noted.

Time	Event
2026-05-01 17:44	Operator session begins; storage migration of esxi02 from HGST to NVMe initiated via robocopy
2026-05-01 19:55	esxi03 robocopy /MOVE complete; F: source cleared
2026-05-01 20:09	esxi02 robocopy complete (E: source preserved as snapshot)
2026-05-01 ~22:00	esxi02 reopened in Workstation from J:; "I Moved It" preserves UUID
2026-05-02 00:13:13	**AI assistant recommends Ensure Accessibility for MM** (msg `01Vte752bDLZjv4q3pqyVjsr`)
2026-05-02 ~00:30	Operator places esxi02 into MM with Ensure Accessibility
2026-05-02 ~00:45	esxi02 reimaged with fresh ESXi 9.0.1 build 24957456; rejoined cluster (no disk groups claimed)
2026-05-02 ~01:00	vSAN compliance rebuild storm begins; resync 188 GB observed
2026-05-02 ~01:30	SDDC Manager and LCM VMs report Orphaned in vCenter inventory
2026-05-02 ~01:35	vCenter VM console shows EXT4 journal abort errors on dm-14
2026-05-02 ~01:45	vCenter services unresponsive; manual power-off
2026-05-02 ~02:00	vSphere Skyline Health reports 9 inaccessible objects
2026-05-02 16:30	Operator initiates host-side recovery via AI assistant + plink
2026-05-02 16:40	DOM ownership abdication executed on all 9 UUIDs (per KB 405527) — no recovery
2026-05-02 16:51	objtool delete executed on 9 dead objects (per KB 428195) — cluster Skyline returns to clean
2026-05-02 ~18:00	VCSA 9.0.1 fresh install commenced on esxi04

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7. Recovery Actions Taken

7.1 Tier 1 — Ownership Abdication (KB 405527)

Goal: Determine whether inaccessibility was due to stuck DOM ownership on a single host (recoverable) versus genuinely missing data (not recoverable).

Procedure: All 9 inaccessible objects' DOM ownership was held by esxi03 (the master post-partition). Per [KB 405527](), executed from esxi03:

```
 # Run on the ESXi host only — vsish is the ESXi VMkernel-sysinfo shell utility
# and is not available on a Windows workstation. SSH into the master host
# (here esxi03) and execute the loop in its local shell.
for u in <list-of-9-uuids>; do
  vsish -e set /vmkModules/vsan/dom/ownerAbdicate "$u"
done
```

Result: All 9 commands returned `rc=0`. Ownership transferred to esxi01 / esxi04 as expected. Object health remained `inaccessible — Lost data availability.(APD)` for all 9. **Conclusion: data is genuinely missing, not ownership-stuck.**

7.2 Tier 3 — Object Cleanup (KB 428195)

Goal: Remove dead object metadata from CMMDS so that the cluster Skyline Health returns to clean and the new vCenter (when deployed) does not see polluted health alarms.

Procedure: Per [KB 428195](), executed from esxi03:

```
 # Run on the ESXi host only — objtool is the ESXi vSAN object-management
# binary at /usr/lib/vmware/osfs/bin/ and is not available on a Windows
# workstation. SSH into the host and execute the loop in its local shell.
# DESTRUCTIVE: per KB 428195 the -f delete is permanent and non-reversible.
for u in <list-of-9-uuids>; do
  /usr/lib/vmware/osfs/bin/objtool delete -u "$u" -f
done
```

> **Warning per KB 428195:** "*Using `objtool delete` will permanently destroy the targeted vSAN object. This action is final and non-reversible; there is no way to restore the object once the command has been issued.*"

Result: All 9 deletes returned `rc=0`. Object health summary moved from 9 inaccessible to 0 inaccessible. 102 healthy objects remained untouched.

7.3 Management Plane Redeploy

With the management plane unrecoverable in place, redeploy was initiated:

1. **VCSA 9.0.1 fresh install** to esxi04 via Host Client OVA deploy (avoids dependency on the broken vCenter).
2. **Re-add hosts esxi01–04 to the new vCenter** post-deploy.
3. **Re-claim esxi02's empty disks** to vSAN via *Cluster → Configure → vSAN → Disk Management → Claim Unused Disks*. Triggers re-resync to the rejoined member.
4. **SDDC Manager redeploy and workload domain reconciliation** — deferred (not required for a working vSphere + vSAN environment; can follow when bandwidth permits).
5. **Apply licenses** — fresh 60-day evaluation on the new vCenter and cluster.

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8. Lessons Learned

1. **Maintenance Mode for any destructive host operation MUST use Full Data Migration.** Ensure Accessibility is acceptable only for transient outages where the host returns with its disks intact within the repair-delay timer (60 min default). Reimage = destructive = Full Data Migration, every time, no exceptions.

2. **The user's own published rebuild plan (VCF9-Host-vCenter-Rebuild-Plan.md) was correct and would have prevented this incident if followed.** The plan exists and cites the right Broadcom KBs. The breakdown was at the moment of operation, not in documentation.

3. **AI assistant guidance must be cross-checked against documented procedures before execution.** When an AI is asked to choose between two options and the correct answer is a third option not listed, the AI must refuse the false binary. Trust without verification on operational steps that mutate storage state is unsafe.

4. **vCenter file-based backup is non-optional.** A 5-minute VAMI configuration would have made recovery a 30-minute restore instead of a multi-hour redeploy with permanent loss of historical inventory state.

5. **Active snapshots on management VMs during storage-impacting operations compound risk.** Always consolidate snapshots before maintenance.

6. **HDD-backed nested vSAN cannot absorb large parallel compliance rebuilds without I/O latency spikes.** This is a physical limit of the lab hardware, not a configuration error — but it makes EA-class events more dangerous in this environment than in production-grade flash-backed vSAN. The mitigation is to never rely on EA for destructive operations in the first place.

7. **SDDC Manager's commission/decommission workflow is a guardrail.** Bypassing it for a host reimage removes the coordinated VCF-state evacuation. Use the prescribed VCF path even when it is slower.

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9. Preventive Measures

9.1 Pre-Rebuild Checklist

Before any ESXi host reimage on a vSAN cluster, all of the following must be true:

- [ ] `esxcli vsan debug resync summary get` reports 0 GB remaining
- [ ] `esxcli vsan debug object health summary get` reports 0 inaccessible, 0 reduced-availability-with-no-rebuild
- [ ] `esxcli vsan cluster get` reports full member count
- [ ] **vCenter file-based backup** has run successfully within the last 24 hours and the backup file is verified
- [ ] **No active snapshots** on management VMs (vCenter, SDDC Manager, LCM, NSX Manager)
- [ ] **VCF Operations is reachable** for post-rebuild license application (KB 428053)
- [ ] **Exact-build ESXi installer ISO** is available (KB 327032)
- [ ] **Operator confirms Maintenance Mode option = Full Data Migration**, not EA, not NDM

9.2 MM Mode Decision Matrix

Scenario	MM Option	Why
Quick reboot (< 60 min total absence)	No Data Migration	Repair delay timer covers the window
Brief outage with disks intact, host returns < 60 min	Ensure Accessibility	Components stay; host returns with same data
Reimage / reinstall ESXi	**Full Data Migration**	**Disks will be destroyed; components must be evacuated first**

Scenario	MM Option	Why
Hardware swap (same disks moved to new chassis)	Ensure Accessibility	If disks return with data, EA is acceptable
Hardware swap (new disks)	Full Data Migration	Disks will not return with data
Decommission permanently	Full Data Migration	Components must be relocated for cluster integrity

9.3 Backup Policy

Effective immediately after the new vCenter deploy is validated:

1. **vCenter VAMI backup** configured to a target *not* on the vSAN being protected. Preferred targets in this environment: SFTP server on the Windows workstation's `I:\VCF-Backups\` directory or an external NAS.
2. **Daily schedule**, retain 7 daily + 4 weekly.
3. **Restore drill** within 30 days to verify the backup is actually usable.
4. **SDDC Manager backup** configured similarly when SDDC Manager is redeployed.
5. **NSX Manager backup** configured similarly per existing NSX backup pattern.

9.4 AI Assistant Guidance Validation

For operational steps that mutate storage, network, or cluster state:

1. AI recommendations must cite a specific Broadcom KB or vendor documentation source.
2. If the AI offers a choice between options and an option not in the user's prompt is the correct one, the AI must call out the third option and refuse the false binary.
3. Cross-check AI guidance against existing internal runbooks (the *VCF 9.0 ESXi Host, vCenter & NSX Manager Rebuild Plan* is the authoritative reference for this environment) before execution.
4. Destructive operations (`objtool delete`, `vsan-related` commands, `clomd` restart, MM entry) require explicit per-step confirmation, even under broad authorization.

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10. References — Broadcom KBs

- KB 327031 — Changing the default repair delay time for a host failure in vSAN
- KB 327032 — How to add a host back to a vSAN cluster after rebuild/reimage
- KB 391111 — Procedure to remove and re-add a vSAN node
- KB 405527 — vSAN Object Inaccessibility Causing VM Orphaned State Due to Ownership on Maintenance Host
- KB 428053 — Cannot apply licenses to ESXi/vCenter 9.0 if licenses already expired and VCF Ops unavailable
- KB 428195 — Resolving Inaccessible vSAN Objects and Orphaned Snapshots using objtool in ESXi 7.x and ESXi 8.x
- KB 397411 — How to investigate and clean Inaccessible VSAN objects
- KB 326473 — Working with inaccessible objects on vSAN cluster
- Internal: *VCF 9.0 ESXi Host, vCenter & NSX Manager Rebuild Plan* (Virtual Control LLC, 2026-04-25)
- Internal: *vSAN Health Check Handbook* (Virtual Control LLC)

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11. Day 2 — Cluster Restoration Activity (2026-05-03)

This section documents the second-day recovery activity to bring the cluster from "fresh vCenter, 3-host vSAN, esxi02 isolated" to "4-host vSAN healthy."

12.1 Day 2 Starting State

After Day 1's redeploy phase, the environment was:

- New vCenter Server 9.0.1 deployed on esxi04 at 192.168.1.69 (fresh install)
- 4 ESXi hosts re-added to the new vCenter at datacenter level
- vSAN sub-cluster healthy at 3 members (esxi01, esxi03, esxi04) with 111 healthy objects, 0 inaccessible
- esxi02 in vCenter inventory but isolated: no vSAN vmkernel adapter, claiming itself as MASTER of a 1-host sub-cluster
- Original `vcenter-cl01-vds01` vDS lost with the old vCenter database; surviving hosts running on orphaned `dvportgroup-22/24/25` references that the new vCenter could not see or manage

12.2 The vDS Loss

vCenter Distributed Switch configuration is stored in the vCenter database, not on the hosts. Hosts cache the dvport state locally and continue forwarding traffic on the cached state, but vCenter has no way to reconstruct a vDS from host-side cache. When the original vCenter was destroyed in the Day 1 cascade, the vDS configuration was lost with it.

PowerCLI inventory query confirmed the new vCenter saw zero distributed switches:

```
 PS> Get-VDSwitch
(returns nothing)

PS> Get-VirtualPortGroup
Name               VirtualSwitch VLanId
----               ------------- ------
VM Network         vSwitch0           0
Management Network vSwitch0           0
vsan-tmp           vSwitch-tmp        0
vmotion-tmp        vSwitch-tmp        0
mgmt-tmp           vSwitch-tmp        0
```

The orphaned `dvportgroup-25` was visible on `Get-VMHostNetworkAdapter` per-host queries (e.g. esxi01 vmk2 PortGroupName: `dvportgroup-25`), but no `Get-VDPortgroup` cmdlet could find it because it was a host-side reference to a vCenter object that no longer existed.

12.3 Decision — vDS Rebuild Versus Coexist

Two paths were available to restore esxi02:

Path	Approach	Risk	Time
Path X — Rebuild vDS	Create new vDS in new vCenter, migrate all 4 hosts and vmkernels off orphan vDS	High — migrating live hosts off the orphan vDS could partition the working 3-host vSAN	1–2 hours, careful
Path Y — Standard vSwitch on esxi02 only	Give esxi02 a standard vSwitch with vmk on vSAN subnet; leave the working 3 hosts untouched	Low — only touches the host that has nothing to lose	~5 minutes

Path Y was chosen. vSAN uses IP-level connectivity at L3 — it does not require all hosts to be on the same logical vDS, only that vmkernels can reach each other on the vSAN VLAN. The mixed end-state (esxi01/03/04 on orphan vDS, esxi02 on standard vSwitch) is functionally fine for vSAN.

12.4 esxi02 vSAN Network Setup (Path Y)

Driven via PowerCLI:

```
 $h = Get-VMHost esxi02.lab.local
$vss = New-VirtualSwitch -VMHost $h -Name 'vSwitch-vsan' -Nic vmnic2
$pg  = New-VirtualPortGroup -VirtualSwitch $vss -Name 'vsan-pg'
$vmk = New-VMHostNetworkAdapter -VMHost $h -PortGroup $pg -VirtualSwitch $vss `
       -IP 192.168.12.124 -SubnetMask 255.255.255.0 `
       -VsanTrafficEnabled $true -Confirm:$false
```

Result: vmk1 created with IP 192.168.12.124, vSAN service tagged. vmnic1 was already in use on vSwitch0; vmnic2 selected as the next free uplink. Connectivity verified by `vmkping -I vmk1 192.168.12.121/.122/.123` — all replies (1–4 ms RTT).

12.5 Unicast Agent Manual Configuration

Network connectivity alone did not cause esxi02 to join the cluster. vSAN 6.6+ uses **unicast mode**: each host maintains an explicit unicast-agent list pointing to its peers. After the rebuild, esxi02 had no peers in its list, and esxi01/03/04 had no entry for esxi02. The cluster could not converge until both sides had each other.

`esxcli vsan cluster join -u <sub-cluster-uuid>` returned `rc=0` but did not actually join the cluster — esxi02 continued to claim it was a 1-host master. The `join` command sets the sub-cluster UUID but does not populate the unicast-agent list.

The supported procedure is to populate unicast agents manually via `esxcli vsan cluster unicastagent add` on every host. Node UUIDs gathered from each host:

Host	Node UUID	vSAN IP
esxi01	`6983ca7b-7444-ee14-a551-000c29fa3cc7`	192.168.12.121
esxi02	`69f5436c-76ba-2b9a-ffcf-000c29649d0a`	192.168.12.124
esxi03	`69eccff6-0f73-3905-5c57-000c29b4cef4`	192.168.12.122
esxi04	`6983dcc7-be7d-49ce-02c4-000c2951493d`	192.168.12.123

Per-host script form:

```
 # Run on the ESXi host only — esxcli and /etc/init.d/* are ESXi-shell
# utilities and are not available on a Windows workstation. SSH into each
# host and execute the matching block locally.
# On esxi01, esxi03, esxi04 — add the rebuilt host:
esxcli vsan cluster unicastagent add -t node \
  -u 69f5436c-76ba-2b9a-ffcf-000c29649d0a \
  -U true -a 192.168.12.124 -p 12321

# On esxi02 — add all three peers (some may report "Already exists"
# if vsanmgmtd auto-populated; benign):
esxcli vsan cluster unicastagent add -t node \
  -u 6983ca7b-7444-ee14-a551-000c29fa3cc7 -U true -a 192.168.12.121 -p 12321
esxcli vsan cluster unicastagent add -t node \
  -u 69eccff6-0f73-3905-5c57-000c29b4cef4 -U true -a 192.168.12.122 -p 12321
esxcli vsan cluster unicastagent add -t node \
  -u 6983dcc7-be7d-49ce-02c4-000c2951493d -U true -a 192.168.12.123 -p 12321

/etc/init.d/vsanmgmtd restart
```

Within ~5 seconds of `vsanmgmtd` restart on esxi02, the cluster converged:

```
 Sub-Cluster Member Count: 4
Sub-Cluster Member HostNames: esxi03, esxi04, esxi01, esxi02
Local Node Health State: HEALTHY
```

12.6 Disk Group Auto-Rediscovery

Surprising and useful finding: **when esxi02 properly rejoined the cluster, vSAN auto-discovered its existing disk groups from on-disk metadata.** No manual `Claim Unused Disks` step was needed.

`Get-VsanDisk -VMHost esxi02.lab.local` returned 4 vSAN disks already claimed.

The mechanism: an ESXi reinstall with "Install ESXi, preserve VMFS datastore" wipes the boot/system partitions but does not touch attached vmdk content. The vSAN disks' on-disk metadata (LSOM headers, disk-group config) survived. When the host reconnected with proper unicast and CMMDS sync, the vSAN cluster replayed the disk-group ownership and the components became accessible.

12.7 Hindsight Lesson — Premature objtool delete

This is the most important lesson from Day 2.

On Day 1, after Tier 1 (ownership abdication, KB 405527) failed to restore accessibility on the 9 objects, `objtool delete -f` (KB 428195) was executed to clean orphan metadata. That action was permanent and irreversible.

Day 2 demonstrated that the data on esxi02's disks was physically intact the whole time. When esxi02 was properly reintegrated with unicast configured, vSAN re-discovered the disk groups and components became valid again. Had this rejoin path been executed *before* the objtool delete, some of the 9 inaccessible objects might have become accessible — meaning the irreversible action that destroyed the original vCenter, SDDC Manager, and LCM metadata may not have been necessary.

Why this wasn't apparent on Day 1:

- Broadcom's documented procedure for inaccessible objects (KB 405527, KB 428195, KB 397411) does not explicitly cover "rebuilt host with on-disk vSAN data still present, just not yet rejoined to cluster." The KBs assume hosts are either healthy and present, or genuinely failed and gone.
- The 1-host phantom sub-cluster on esxi02 obscured the recoverability — `esxcli vsan cluster get` showed it was master of itself, suggesting it had no useful data.
- Attempts to add esxi02 to a vDS that no longer existed in vCenter looked like an unrelated failure path.
- The cascade pressure (vCenter dead, no backup, multiple management VMs orphaned) created urgency that worked against patience.

Corrective control going forward: whenever an ESXi host has been rebuilt and joined a cluster with inaccessible vSAN objects, **before any destructive action (objtool delete, force-purge), verify the rebuilt host's vSAN networking and unicast configuration are both correct, then check object health again.** Rejoin can resurrect inaccessibility caused purely by host-isolation, even when the host's vSAN partitions appear to have been preserved.

A documented "host rejoin checklist before any destructive vSAN cleanup":

1. vSAN vmkernel adapter exists on the rebuilt host (vmk with `Tags: VSAN`)
2. `vmkping` from rebuilt host to *every* peer's vSAN IP succeeds
3. Unicast agent list on the rebuilt host contains every peer (`esxcli vsan cluster unicastagent list`)
4. Unicast agent list on every peer contains the rebuilt host
5. `esxcli vsan cluster get` reports the rebuilt host as a member of the same sub-cluster UUID as peers, with member count matching reality
6. **Then** re-check `esxcli vsan debug object health summary get` — only proceed with destructive cleanup on objects that remain inaccessible

12.8 Day 2 Final State

Metric	Value
Cluster member count	4 (esxi01, esxi02, esxi03, esxi04)
All hosts vSAN health	HEALTHY
Object health	111 healthy, 0 inaccessible, 0 reduced-availability
Resync activity	0 GB / 0 objects
New vCenter	up at 192.168.1.69 (fresh deploy, 60-day eval)
SDDC Manager / LCM / NSX-T Manager	absent — must be redeployed in a separate exercise when needed
vDS state	Mixed: esxi01/03/04 on orphan `vcenter-cl01-vds01` (host-side cache), esxi02 on standard `vSwitch-vsan`

The mixed vDS state is operationally fine for current vSAN traffic but should be unified in a planned, separate maintenance window — see Section 12.

---

12. Future Work — vDS Rebuild (largely completed Day 2)

The vDS rebuild was substantially executed on 2026-05-03 (see Section 11). Sections 12.1–12.5 below document the *original* future-work plan as written before the work began; Section 14 documents the multi-session roadmap for the remaining 5% of work plus broader recovery items.

12.1 Original problem statement

The orphan vDS `vcenter-cl01-vds01` was a known long-term liability after Day 1:

- vCenter could not manage the dvportgroups, so port-group changes (VLAN, MTU, security policies, traffic shaping) could not be made in the UI
- New port groups could not be created on the vDS for new traffic types
- A subsequent host reboot might cause unicast advertisements to drop the dvport state
- Mixed vSwitch / vDS state across hosts is officially unsupported

12.2 Status after Day 2

A new vDS `vcenter-cl01-vds01-new` was created in the new vCenter (MTU 1700, 2 uplinks) with port groups `pg-mgmt`, `pg-vmotion`, `pg-vsan`, `pg-vm-mgmt`, `pg-overlay-tep`. Host migration completion:

Host	vSAN/vMotion vmks	Mgmt vmk	VM NICs	vmnic0 (orphan uplink)
esxi01	new vDS ✓	vSwitch-tmp (interim)	n/a	moved to new vDS
esxi02	new vDS ✓	new vDS ✓	(no critical VMs)	moved to new vDS
esxi03	new vDS ✓	new vDS ✓	collector ✓	moved to new vDS

Host	vSAN/vMotion vmks	Mgmt vmk	VM NICs	vmnic0 (orphan uplink)
esxi04	new vDS ✓	new vDS ✓	vcf-ops ✓, nsx-manager ✓, **vcenter still on orphan**	**still on orphan (carries vcenter)**

12.3 Remaining loose ends

- vcenter VM NIC still on orphan `dvportgroup-23`
- esxi04 vmnic0 still attached to orphan vDS (keeps vcenter alive)
- esxi01 vmk0 (mgmt) still on `vSwitch-tmp` (intentional safety during migration)
- NSX vmks (`vmk10` , `vmk11` , `vmk50`) on each of esxi01/03/04 are on orphan dvportgroups, isolated from physical paths — NSX overlay non-functional on those hosts
- Orphan vDS proxy switches still present in host-side cache

12.4 The blocking issue: vCenter VM NIC migration

PowerCLI `Set-NetworkAdapter` and `ReconfigVM_Task` both fail when migrating a VM's NIC FROM an orphan dvportgroup that vCenter cannot validate, with error "*The object or item referred to could not be found.*" This applies to vCenter's own VM as well — chicken-and-egg.

The Host Client at `https://<host>/ui/` accepts NIC edits but ONLY when the *target* port group has **ephemeral** binding, not the default static binding. This blocks direct migration to `pg-vm-mgmt` (which is non-ephemeral).

A `.vmx` file edit (powered-off direct edit on the host) was attempted on 2026-05-03 — the new switchId/portgroupId were applied but the vNIC came up with `NO-CARRIER` because the dvport binding did not establish at boot. The .vmx was reverted from backup; vCenter restored to running state on the orphan port group.

12.5 Solution path documented for the next session

The supported workaround that was identified but not yet executed: **ephemeral port group as a stepping stone.**

1. Create `pg-vm-mgmt-eph` on `vcenter-cl01-vds01-new` with **ephemeral port binding**: `powershell New-VDPortgroup -VDSwitch 'vcenter-cl01-vds01-new' -Name 'pg-vm-mgmt-eph' -PortBinding Ephemeral`
2. Power off vcenter VM
3. Use Host Client at `https://192.168.1.82/ui/` to edit vcenter NIC → assign to `pg-vm-mgmt-eph` (Host Client allows ephemeral PGs — this is the bridge)
4. Power on vcenter, wait ~15 min for services
5. Once vCenter UI is up: migrate the NIC from `pg-vm-mgmt-eph` to `pg-vm-mgmt` via UI — both are on the new vDS, no orphan reference to validate
6. Move esxi04 vmnic0 from orphan to new vDS
7. Migrate esxi01 vmk0 from vSwitch-tmp to pg-mgmt; remove vSwitch-tmp
8. Delete `pg-vm-mgmt-eph`
9. Per-host: remove orphan vDS proxy switch from Host → Configure → Networking

Reference: this approach is documented in Section 14.2 (Session 2) of the recovery roadmap.

---

13. Recovery Roadmap (Multi-Session)

This roadmap defines the remaining work to bring the environment back to fully clean steady state — including replacing the lost VCF management plane (SDDC Manager + LCM) and the storage migration originally planned in `VCF9-Host-vCenter-Rebuild-Plan.md`.

Hard rules carried forward into every session:

- No more multi-session work in one go — each phase ends with a verified-green cluster + a stop
- Backups before any destructive action (Session 5 must precede Sessions 2–4 if at all possible)
- Full Data Migration only for any host MM (the lesson from 2026-05-01)
- Don't bypass SDDC Manager workflows once SDDC Manager is back
- Test in sandbox or rehearse for never-before-tried paths

13.1 Session 1 — 2026-05-03 (closing the day)

Goal: verify cluster green, document state, stop.

Action	Status
vCenter UI + SOAP responsive	✓
All 4 hosts Connected, vSAN healthy, no resync	✓
Stale alarms acknowledged or reset	optional today
RCA v3 written	✓ this update

13.2 Session 2 — Finish vDS migration — **COMPLETED 2026-05-03**

Goal: clean single-vDS topology across all 4 hosts; orphan vDS retired.

13.2.1 Execution

Migration ran in five phases via PowerCLI against the live vCenter — no SDDC Manager workflow available because SDDC Manager was destroyed in the cascade. New vDS `vcenter-cl01-vds01-new` (UUID `50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8`, MTU 1700) was created with two uplinks (`dvUplink1`, `dvUplink2` — the PowerCLI default for `New-VDSwitch`, not `uplink1`/`uplink2` as the orphan vDS used).

Phase	Hosts moved	Method
1	esxi01	`Add-VDSwitchVMHost` then `Add-VDSwitchPhysicalNetworkAdapter` (one uplink at a time, no LACP)
2	esxi02	same
3	esxi03	same
4	esxi04 vmnics — done last because vCenter VM lived here	`pg-vm-mgmt-eph` ephemeral bridge, vCenter graceful power-off, NIC swap via Host Client, power-on, then UI migration to static-binding `pg-vm-mgmt`
5	Cleanup	Removed orphan vDS proxy switches per-host, deleted `pg-vm-mgmt-eph`, removed `vSwitch-tmp` on esxi01

13.2.2 Issues encountered and resolutions

Issue	Resolution
Host Client refused to set vCenter NIC port to a static-binding pg with no free port	Created ephemeral-binding `pg-vm-mgmt-eph` as a stepping stone — ephemeral allocates a port on connect
After editing `.vmx` to remove `ethernet0.portId` the NIC came up `NO-CARRIER`	Connected via ephemeral PG instead — port allocates on demand, no manual portId required

Issue	Resolution
vCenter "object not found" errors during NIC migration	Same ephemeral-PG bridge fixed it — orphan dvport refs were stale

13.3 Session 3 — NSX rebuild — **COMPLETED 2026-05-03**

Goal: NSX foundation restored — Compute Manager registered, hosts re-prepped on the new vDS, TEPs reachable.

13.3.1 vCenter cert chain fix (prerequisite — KB 322036 / 372076)

NSX Compute Manager re-registration failed with `error_code: 90348 — Failed to import trusted root certificate`. Root cause: vCenter's `MACHINE_SSL_CERT` VECS entry contained only the leaf certificate; the VMCA root was in `TRUSTED_ROOTS` but not appended to what vpxd serves. NSX 9 requires the leaf+CA chain in a single response.

Repair on VCSA bash shell:

```
# Run on the appliance only — vecs-cli (/usr/lib/vmware-vmafd/bin/) and
# service-control are VCSA-internal binaries and are not exposed on a
# Windows workstation. SSH to the VCSA as root, drop into the bash shell
# (Command> shell), and execute these in the appliance's local shell.
# Backup
mkdir -p /root/cert-backup-$(date +%Y%m%d-%H%M%S)
cd /root/cert-backup-*

# Export current leaf + key
/usr/lib/vmware-vmafd/bin/vecs-cli entry getcert --store MACHINE_SSL_CERT --alias __MACHINE_CERT --output
leaf.crt
/usr/lib/vmware-vmafd/bin/vecs-cli entry getkey  --store MACHINE_SSL_CERT --alias __MACHINE_CERT --output
leaf.key

# Export the VMCA root (alias visible via 'entry list --store TRUSTED_ROOTS')
/usr/lib/vmware-vmafd/bin/vecs-cli entry getcert --store TRUSTED_ROOTS \
   --alias c0044838f24b889d6666d9a2337a0e557cd0ceae --output ca.crt

# Concatenate leaf + CA → chain.crt
cat leaf.crt ca.crt > chain.crt

# Replace MACHINE_SSL_CERT entry
/usr/lib/vmware-vmafd/bin/vecs-cli entry delete --store MACHINE_SSL_CERT --alias __MACHINE_CERT -y
/usr/lib/vmware-vmafd/bin/vecs-cli entry create --store MACHINE_SSL_CERT --alias __MACHINE_CERT \
   --cert chain.crt --key leaf.key

# Restart all services
service-control --stop --all
service-control --start --all
```

After service start, vCenter served leaf + VMCA root in chain (verifiable with `openssl s_client -connect <vc>:443 -showcerts | grep -c BEGIN`). NSX CM registration then succeeded.

VMCA root SHA-256 thumbprint for thumbprint-based registration: `DA:1F:F8:B3:3A:80:7E:61:33:06:DD:7C:7E:C7:40:B8:B5:B4:72:D3:14:98:C2:6C:0F:83:BE:A7:F6:9B:D9:38`

13.3.2 Compute Manager registration

POST `https://nsx/api/v1/fabric/compute-managers` with the new vCenter FQDN, `vsphere_local_user_credential`, the leaf-cert thumbprint, and `create_service_account: true`. Result: status `REGISTERED`, connection `UP`, id `5b94798e-babf-4831-a303-e27c95e9f80c`.

13.3.3 Transport Node Profile rebuild (KB-style policy API)

Existing TNP `vcenter-cl01-tnp-01` (id `686cb083-99c9-4b43-b2fb-d961b62bf614`) still pointed at the orphan vDS. Updated via policy API:

```
PUT /policy/api/v1/infra/host-transport-node-profiles/vcenter-cl01-tnp-01
```

Critical body fields:

Field	Value	Note
`host_switch_name`	`vcenter-cl01-vds01-new`	new vDS display_name
`host_switch_id`	`50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8`	new vDS UUID
`host_switch_profile_ids[].value`	`/infra/host-switch-profiles/center-cl01-up-host01`	**id is `center-cl01-up-host01` (no leading "v") — display_name and id diverge**
`uplinks[].vds_uplink_name`	`dvUplink1`, `dvUplink2`	**PowerCLI new-vDS default; orphan was `uplink1` / `uplink2`**
`ip_pool_id`	`/infra/ip-pools/vcenter-cl01-tep-pool`	reused existing TEP pool
`transport_zone_endpoints[].transport_zone_id`	`/infra/sites/default/enforcement-points/default/transport-zones/1b3a2f36-bfd1-443e-a0f6-4de01abc963e`	nsx-overlay-transportzone
`_revision`	`0`	required for PUT

Errors hit and corrected during the PUT:

Error	Cause	Fix
`9536 — uplink uplink1 not found on VDS`	wrong uplink name	use `dvUplink1` / `dvUplink2`
`600 — host-switch-profile not found`	wrong profile id (`vcenter-cl01-up-host01`)	actual id is `center-cl01-up-host01`
`500071 — version mismatch`	missing `_revision`	added `_revision: 0`

13.3.4 Transport Node Collection — apply TNP to cluster

```
POST /api/v1/transport-node-collections
{
  "compute_collection_id": "5b94798e-babf-4831-a303-e27c95e9f80c:domain-c10",
  "transport_node_profile_id": "686cb083-99c9-4b43-b2fb-d961b62bf614",
  "display_name": "vcenter-cl01-tnc"
}
```

Realization SUCCESS in ~30 seconds (hosts already had TN bits from the orphan TNP and just got re-linked to the new vDS — no full uninstall/reinstall).

13.3.5 Verification

Check	Result
All 4 host TNs	state=success, status=UP

Check	Result
TEP IP allocations	8 IPs assigned (192.168.13.101–.108), 2 per host
TEP-TEP fabric	`vmkping -S vxlan -I vmk10 192.168.13.108 -c 3 -s 1572` from esxi01 → 0% loss, jumbo MTU 1700 OK
Tunnel count	0 — **expected** with no overlay segments + no logical ports realized on hosts (NSX 9 lazy-realizes segments)

Tunnels-equal-zero in this state is **not a failure** — NSX 9 does not maintain BFD tunnels between TEPs unless there is at least one logical port realized on a shared overlay segment. The fabric was validated end-to-end via vmkping; tunnels will light up the moment the first workload VM is attached to an overlay segment.

13.3.6 What is NOT yet rebuilt

T0 / T1 / segments / Edge cluster were lost with SDDC Manager and not recreated in this session. They were idle in the pre-incident topology (no tenant workloads were on overlays) and rebuilding them was deferred until VCFA bring-up needs them.

13.4 Session 4 — VCF lifecycle redeploy + brownfield convergence — **COMPLETED 2026-05-04 ~20:24 local**

Goal: SDDC Manager + Fleet/LCM back online; existing vSphere+NSX+cluster converged into VCF management via the VCF Installer brownfield path.

13.4.1 Standalone OVA deploys

Skipped Cloud Builder; went the standalone-OVA path because the underlying cluster was already healthy. Deployed both via `ovftool` against vCenter, targeting specific hosts to balance load:

Appliance	OVA	Target host	IP / FQDN	RAM/CPU	Outcome
Fleet/LCM	`VCF-OPS-Lifecycle-Manager-Appliance-9.0.1.0.24960371.ova`	esxi01	192.168.1.95 / lcm.lab.local	12 GB / 4 vCPU	Deployed; UI title self-identifies as "VMware Cloud Foundation Fleet Management Appliance" — direct UI access intentionally disabled in 9.0+, managed via vcf-ops
SDDC Manager (Installer)	`VCF-SDDC-Manager-Appliance-9.0.1.0.24962180.ova`	esxi04	192.168.1.241 / sddc-manager.lab.local	16 GB / 4 vCPU, ~914 GB thin disks	Deployed; first-boot redirects to **VCF Installer UI** at `/vcf-installer-ui` — the appliance promotes itself to SDDC Manager after convergence completes

ovftool deploy parameters injected via `--prop` for both: `vami.ip0`, `netmask0`, `gateway`, `domain`, `searchpath`, `DNS` (uppercase for SDDC Manager, lowercase `dns` for LCM), `ROOT_PASSWORD`, hostname, NTP. Disk mode `thin`, datastore `vcenter-cl01-ds-vsan01`.

Naming gotcha discovered: the OVA filename `VCF-OPS-Lifecycle-Manager-Appliance` (and depot folder `VRSLCM`) is legacy carry-over from vRealize Suite Lifecycle Manager. In VCF 9.0+ this same OVA serves the **Fleet Management Appliance** role. There is no separate Fleet OVA. If a pre-9.0 environment had both `lcm` and `fleet` VMs (as this lab did at 192.168.1.95 and .78), the .78 instance is redundant in 9.0+ and was not redeployed.

13.4.2 Cluster EVC enablement (prerequisite for VM mobility)

The VCF Installer requires DRS fully automated, which in turn requires VM mobility across hosts. Initial vMotion attempts failed with `Module CPUID power on failed` — diagnosed as nested-ESXi CPU feature exposure mismatch (all 4 hosts on the same Xeon Gold 6140 physical CPU but exposing different CPUID flags to nested guests).

Set cluster EVC to `intel-skylake` (matches 6140 generation) via:

```
$cl = Get-View (Get-Cluster -Name 'vcenter-cl01')
$evcMgr = Get-View $cl.EvcManager()
$evcMgr.ConfigureEvcMode_Task('intel-skylake', $null)
```

PowerCLI 13.3.0 doesn't expose `Set-Cluster -EVCMode`; had to call the SDK method directly via Get-View. After EVC enabled, **VMs running pre-EVC kept their old CPU exposure until power-cycled**. Procedure:

1. Hard-stopped vCLS pair (vCenter auto-recreates them under the new EVC mask within ~3 min)
2. Graceful shutdown of vCenter, NSX, vcf-ops via VAMI / vim-cmd, then power-on under EVC
3. Verified each VM's `Runtime.MinRequiredEVCModeKey` ≤ skylake post-restart
4. Test vMotion succeeded after cycle

13.4.3 Pre-validation remediation cycle

VCF Installer's "Existing Components" validation surfaced 9 distinct findings on first run. Fixed each before re-running:

Finding	Fix
esxi02, esxi03 NTP not configured (set to `127.0.0.1 noselect`, ntpd not running)	`esxcli system ntp set --server=192.168.1.230 --enabled=true; /etc/init.d/ntpd restart`
esxi02, esxi03 had leftover standard `vSwitch0` with `Management Network` / `VM Network` PGs (no vmks attached)	Removed via PowerCLI: delete portgroups then `Remove-VirtualSwitch`
Cluster DRS not fully automated	`Set-Cluster -DrsEnabled $true -DrsAutomationLevel FullyAutomated`
Cluster upgrade policy "Evacuate Offline VMs" did not match SDDC Manager default	UI: cluster → Updates tab → Image → "..." → Edit Cluster Remediation Settings → check **Migrate powered off and suspended VMs to other hosts in the cluster, if a host must enter maintenance mode**
NSX VIP not configured (0.0.0.0)	`POST /api/v1/cluster/api-virtual-ip?action=set_virtual_ip&ip_address=192.168.1.70` (query-param API, not body — body returns error 255)
`clusterAdvancedVsanConfigInSyncStatus` failure: esxi01/03/04 had `VSAN.ClomMaxComponentSizeGB=100` and `ClomRebalanceThreshold=50`; esxi02 had defaults `255` / `80` (rebuilt from scratch during recovery)	Set all 4 to Skyline Health-recommended `180` / `80` (cluster's smallest disk is 100 GB so 180 is the recommended max component cap)
esxi02 vMotion vmknic missing	Diagnosed: the `vmotion` netstack didn't exist on esxi02 post-rebuild. Created via `esxcli network ip netstack add --netstack=vmotion`, then added vmk on dvport 131 in `pg-vmotion`: `esxcli network ip interface add --interface-name=vmk2 --netstack=vmotion --dvs-name='vcenter-cl01-vds01-new' --dvport-id=131` + `ipv4 set --type=static --ipv4=192.168.11.124 --netmask=255.255.255.0`
vSAN Disk Balance — "Disk rebalance is needed but disabled"	Cluster vSAN Advanced Options: toggle **Automatic rebalance ON** (background data move started; 12 objects / 61 GB / ~2 hours)

Finding	Fix
`lcm.lab.local` DNS A record missing (NXDOMAIN) — initially had been swapped to `fleet.lab.local` per a misguided suggestion	Re-added `lcm` → 192.168.1.95 forward + PTR records on Windows DNS at .230. Both `lcm.lab.local` and `fleet.lab.local` resolve to the same VM

13.4.4 Brownfield convergence wizard

VCF Installer wizard at `https://sddc-manager.lab.local`:

1. Landing page: "Online depot connection active ✓" — selected "Deployment Wizard"
2. **What would you like to deploy?**: chose **"Deploy a VCF Instance in an existing VCF fleet"** (right option for brownfield — uses existing fleet appliance + adds new VCF Instance with management domain wrapping existing infra)
3. **Existing Components**: `VMware vCenter: Yes`, `NSX Manager: Yes` (declares the existing infra as reusable)
4. **General Information**: Version 9.0.1.0, VCF Instance Name `mgmt-domain-01`, Management domain name `mgmt`, Deployment Model **Simple (Single-node)** (homelab — saves ~24 GB RAM vs HA 3-node)
5. **VCF Operations**: auto-detected `vcf-ops.lab.local` (green ✓), Fleet Management `lcm.lab.local`, Operations Collector `collector.lab.local` (existing, powered off — wizard's plan was to redeploy)
6. **vCenter**: existing — `vcenter.lab.local`, root + SSO creds, NSX `nsx-manager.lab.local` (wizard auto-detected NSX hostname as `nsx-node1` initially; corrected to `nsx-manager` since that's what's in DNS), Edge Cluster sync **disabled** (no Edges)
7. **SDDC Manager**: `sddc-manager.lab.local`, "VCF Installer Deployment: Re-used for SDDC Manager" — the .241 appliance promotes itself
8. **VCF Automation**: showed **Missing** — wizard does not deploy VCFA in this flow; warning to deploy VCFA later via vcf-ops
9. **Validate & Deploy**: hit several validation issues all addressed above

13.4.5 Binary downloads (online depot)

Required bundles auto-downloaded from Broadcom's online depot once activated in VCF Installer Settings:

Bundle	Build	Size	Status
SDDC Manager	9.0.1.0.24962180	2.0 GB	✓ Downloaded
Operations Collector (cloud-proxy)	9.0.1.0.24960349	2.8 GB	✓ Downloaded
VCF Automation (VRA)	9.0.1.0.24965341	24.2 GB	✓ Downloaded (~2 hour at ~24 Mbps)
Fleet Management	9.0.1.0.24960371	1.5 GB	Skipped — already running at .95

Three 9.0.2.0 downloads accidentally selected in initial pass; cancelled correctly to stay on 9.0.1 stack.

13.4.6 Convergence outcome

Brownfield convergence wizard completed at **2026-05-04 ~20:24 local time**, reporting 100% complete with all four steps green:

1. ✓ Convert the existing vCenter with existing NSX to a new VCF instance
2. ✓ Configure the appliance as SDDC Manager
3. ✓ Join the existing fleet management appliance
4. ✓ Join the existing operations appliance

Result: a complete VCF-managed environment wrapping the disaster-recovered vCenter+NSX+vSAN cluster. The .241 Installer appliance promoted itself to SDDC Manager. Existing vcf-ops and lcm appliances joined as fleet members.

Next steps from the wizard's "Next Steps" panel: 1. Log in to VCF Operations UI (`vcf-ops.lab.local` , admin) 2. Apply VCF license keys within the 90-day evaluation period

13.4.7 Remaining work

- License keys (90-day eval started 2026-05-04)
- VCFA deploy via vcf-ops Fleet Management → Lifecycle → VCF Management (separate workflow, ~2-3 hours)
- Storage migration — esxi01 to 2 TB NVMe (drive on hand) and Windows C: to 4 TB NVMe (arriving ~2026-05-07) tracked in `project_esxi01_nvme_migration.md`

13.5 Session 5 — Backups & governance (mandatory before further risk) — COMPLETED 2026-05-03

Goal: all management plane components have working file-based backup; restore drill scheduled.

13.5.1 SFTP target setup on Windows workstation (192.168.1.160)

The backup target was the workstation's `I:\VCF-Depot\` USB SSD. Steps performed:

1. **Installed Windows OpenSSH Server** via elevated PowerShell: `powershell Add-WindowsCapability -Online -Name 'OpenSSH.Server~~~~0.0.1.0' Start-Service sshd Set-Service sshd -StartupType Automatic` Firewall rule for TCP 22 inbound auto-created.

2. **Created dedicated `admin` local user** with a strong password (stored in credential vault — see `reference_homelab_esxi_creds`), password-never-expires, member of `Users` group. Reused existing local account, just reset password and ensured group membership.

3. **Granted `admin` Modify ACL** on `I:\VCF-Depot\backups` (recursive): `powershell icacls 'I:\VCF-Depot\backups' /grant 'admin:(OI)(CI)M' /T`

4. **Configured sshd_config chroot** so `admin`'s SFTP root is the backups directory — both NSX and vCenter use clean Linux-style paths without the `:` colon that NSX rejects: `Match User admin ChrootDirectory I:\VCF-Depot\backups ForceCommand internal-sftp AllowTcpForwarding no X11Forwarding no` Restarted sshd. After chroot, admin's SFTP `/` shows `vcenter/` and `nsx-manager/` directly.

5. **Created subdirectories**: `I:\VCF-Depot\backups\vcenter\` and `I:\VCF-Depot\backups\nsx-manager\` .

13.5.2 Configuration values

vCenter VAMI Backup (`https://192.168.1.69:5480` → Backup → Configure):

Field	Value
Backup location	`sftp://192.168.1.160:22/vcenter`
Username	`admin`
Password	(from credential vault — see `reference_homelab_esxi_creds`)
Encryption password	(set by operator, documented)
Schedule	Daily

Field	Value
Retention	10 backups
Additional data	Stats, Events, and Tasks

NSX Manager Backup (`https://192.168.1.71` → System → Backup & Restore):

Field	Value
FQDN/IP	`192.168.1.160`
Port	`22`
Protocol	SFTP
Directory	`/nsx-manager`
Username	`admin`
Password	(from credential vault — see `reference_homelab_esxi_creds`)
SSH Fingerprint	**`SHA256:d9c+ICSL5rOMGns/3ig8YrcRnBCKE7zjD+ZlIezPkx8`** (ECDSA)
Backup passphrase	(set by operator, documented)

13.5.3 NSX SFTP fingerprint format gotcha

NSX Manager's SFTP backup config validates the SSH fingerprint against the live server. **NSX rejects RSA-SHA256, RSA-MD5, ED25519-SHA256, ED25519-MD5** with `Error code: 29004 — Configured fingerprint did not match fileserver`. Only the **ECDSA SHA256** format with `SHA256:` prefix was accepted.

To compute the right fingerprint for any SFTP server:

```
# Workstation-runnable (Git Bash / MSYS) — calls Windows OpenSSH binaries
# directly. Translates 1:1 to the PowerShell sibling block below; pick whichever
# shell is open. Pipe the keyscan output through ssh-keygen -l to print the
# fingerprint of every host key the server advertises; copy the ECDSA SHA256
# line into the NSX backup config.
"/c/Windows/System32/OpenSSH/ssh-keyscan.exe" -t rsa,ecdsa,ed25519 <ip> > /tmp/keys.txt
"/c/Windows/System32/OpenSSH/ssh-keygen.exe" -l -f /tmp/keys.txt
# Use the ECDSA entry's SHA256 fingerprint in NSX
```

```
# PowerShell sibling — compute SFTP server fingerprints from a Windows shell.
# Calls the same Windows OpenSSH binaries; no MSYS/Git Bash dependency.
# Replace <ip> with the SFTP server (e.g. 192.168.1.160). The ECDSA SHA256
# line is the one NSX will accept in System -> Backup & Restore.
$sftpHost = '<ip>'
$keysFile = Join-Path $env:TEMP 'sftp-keys.txt'
& "$env:WINDIR\System32\OpenSSH\ssh-keyscan.exe" -t rsa,ecdsa,ed25519 $sftpHost | Set-Content -Path
$keysFile -Encoding ascii
& "$env:WINDIR\System32\OpenSSH\ssh-keygen.exe" -l -f $keysFile
# Copy the ECDSA SHA256:... value from the output into NSX's "SSH Fingerprint" field.
```

13.5.4 Initial backup verification

Backup	Size on disk	Path	Status
vCenter (manual)	446 MB	`I:\VCF-Depot\backups\vcenter\vCenter\sn_vcenter.lab.local\...`	✓ Successful

Backup	Size on disk	Path	Status
NSX (manual)	94 MB	`I:\VCF-Depot\backups\nsx-manager\cluster-node-backups\...`	✓ Successful

13.5.5 Outstanding governance items (not yet done)

- SDDC Manager backup target — pending Session 4 (SDDC Manager redeploy)
- Restore drill in sandbox within 30 days
- Off-host copy of backups (USB SSD on workstation is single-point-of-failure for the backups themselves)

13.6 Session 6 — Storage migration completion (parallel to others when NVMe arrives)

Goal: original Phase 2 of `VCF9-Host-vCenter-Rebuild-Plan.md` completed — all 4 hosts on NVMe.

1. New NVMe drive (arriving 2026-05-03) replaces D: backing esxi01
2. Workstation-level move: power off esxi01, robocopy disks to new NVMe, "I Moved It" on power-on
3. Per Maintenance Mode = **Full Data Migration only** (the lesson from 2026-05-01)
4. Once esxi01 is on NVMe, retire E: HGST and F: SATA per the original plan
5. All 4 hosts on NVMe; original Phase 2 storage migration complete

13.7 Session Dependencies

```
Session 1 ✓ (2026-05-03)
     │
     ├──► Session 5 ✓ (BACKUPS — completed 2026-05-03)
     │        │
     │        ├──► Session 2 ✓ (vDS finish — completed 2026-05-03)
     │        │        │
     │        │        └──► Session 3 ✓ (NSX rebuild — completed 2026-05-03)
     │        │                 │
     │        │                 └──► Session 4 ✓ (VCF redeploy — COMPLETED 2026-05-04 20:24: SDDC Manager +
Fleet deployed, EVC + 9 remediations, brownfield convergence 100%)
     │        │
     │        └──► Session 6 ◯ (storage swap — esxi01 → 2 TB NVMe today, Windows C: → 4 TB NVMe 2026-05-07)
     │
```

---

14. Document Information

Item	Value
Document title	VCF 9.0 Management Plane Loss — Root Cause Analysis
Document version	4.1
Author	Virtual Control LLC
Date prepared	2026-05-02 (v1), 2026-05-03 (v2 — added Day 2 recovery + hindsight lesson), 2026-05-03 (v3 — Sessions 2 + 3 executed; vCenter cert chain fix; NSX TNP/TNC rebuild), 2026-05-04 (v4 — Session 4: SDDC Manager + Fleet OVA deploy; cluster EVC; 9 pre-validation remediations; brownfield convergence wizard execution), 2026-05-08 (v4.1 — Option B PowerShell treatment)

Item	Value
Incident date	2026-05-01 cascade; 2026-05-02/03 recovery
Environment	VCF 9.0.1 nested lab — 4-host vSAN OSA cluster
Severity	P1 — Total management plane loss
Recovery type	Full redeploy of management plane
Classification	Internal — Operational
Distribution	Engineering, Lab Operations
Related documents	VCF9-Host-vCenter-Rebuild-Plan.md, vSAN-Health-Check-Handbook.md, VCF-vSAN-Cluster-Partition-Recovery.md
Source session	AI assistant session `46f1b37d-9f39-41ea-b592-32cd05b616f8` ; bad-recommendation msg `01Vte752bDLZjv4q3pqyVjsr` at 2026-05-02 00:13:13 UTC

Revision History

Version	Date	Notes
1.0	2026-05-02	Initial RCA covering the 2026-05-01 cascade (MM/EA decision, journal abort, Tier 1/2 recovery actions, redeploy outcome).
2.0	2026-05-03	Added Day 2 (Section 11) — esxi02 unicast-agent rejoin, disk-group auto-rediscovery, hindsight lesson on premature `objtool delete` .
3.0	2026-05-03	Sessions 2 + 3 executed — vDS rebuild via ephemeral-PG bridge, vCenter `MACHINE_SSL_CERT` chain fix (KB 322036/372076), NSX Compute Manager re-registration, TNP/TNC rebuild on the new vDS.
4.0	2026-05-04	Session 4 — standalone OVA deploys of Fleet/LCM and SDDC Manager Installer; cluster EVC `intel-skylake` ; 9 pre-validation remediations; brownfield convergence wizard 100% green at 20:24 local.
4.1	2026-05-08	Option B PowerShell treatment: added top-of-doc callout linking to `../POWERSHELL-TRANSLATION-REFERENCE.md` ; added a paste-ready PowerShell sibling for the workstation-runnable Section 13.5.3 ssh-keyscan / ssh-keygen ECDSA-fingerprint block (uses `$env:WINDIR\System32\OpenSSH` binaries with no MSYS dependency). The four ESXi-shell / appliance-shell blocks (Section 7.1 `vsish ownerAbdicate` , Section 7.2 `objtool delete` , Section 12.5 `esxcli vsan cluster unicastagent` + `/etc/init.d/vsanmgmtd restart` , Section 13.3.1 `vecs-cli` + `service-control`) were flagged "Run on the ESXi host only" / "Run on the appliance only" with no PS sibling — they are not workstation-translatable. No content/procedure changes; no PS 5.1 cert-bypass needed (no PS sibling makes a REST call in this document).